Definite Integration: Questions and Shortened Solutions

Question 01

English

Let $p : \mathbb{R} \to \mathbb{R}$ be a polynomial function of degree 3 such that $p(0) = 12$ and

$$\int_0^1 p(t)\,dt = \int_0^2 p(t)\,dt = \int_0^3 p(t)\,dt = 0.$$

Then $\displaystyle\int_4^5 p(t)\,dt$ is equal to

(A) 192
(B) -96
(C) -192
(D) -48

Only one option is correct.

हिन्दी

माना कि $p : \mathbb{R} \to \mathbb{R}$ घात 3 का एक बहुपद फलन है जिसके लिए $p(0) = 12$ तथा

$$\int_0^1 p(t)\,dt = \int_0^2 p(t)\,dt = \int_0^3 p(t)\,dt = 0.$$

तब $\displaystyle\int_4^5 p(t)\,dt$ बराबर है

(A) 192
(B) -96
(C) -192
(D) -48

केवल एक विकल्प सही है।

Solution

Let $P(x) = \int_0^x p(t)\,dt$: a polynomial of degree 4 with $P' = p$ (FTC part 1)
$P(0) = 0$ and the given conditions say $P(1) = P(2) = P(3) = 0$, so $P(x) = kx(x-1)(x-2)(x-3)$
$p(0) = P'(0) = k(-1)(-2)(-3) = -6k = 12 \Rightarrow k = -2$
By FTC part 2, $\int_4^5 p = P(5) - P(4) = k(120 - 24) = 96k = -192$

Correct Option: (C)

Question 02

English

The value of the integral

$$\int_{\pi/2}^{3\pi/2} \frac{(x + \sin x)\,|\cos x|}{|\sin x| + |\cos x|}\,dx$$

is equal to

(A) $\frac{\pi^2}{2}$

(B) $\frac{\pi^2}{4}$

(C) π^2

(D) $\frac{\pi}{2}$

Only one option is correct.

हिन्दी

समाकल

$$\int_{\pi/2}^{3\pi/2} \frac{(x+\sin x)\,|\cos x|}{|\sin x|+|\cos x|}\,dx$$

का मान बराबर है

(A) $\frac{\pi^2}{2}$

(B) $\frac{\pi^2}{4}$

(C) π^2

(D) $\frac{\pi}{2}$

केवल एक विकल्प सही है।

Solution

Let $g(x) = \frac{|\cos x|}{|\sin x|+|\cos x|}$; $\sin(2\pi - x) = -\sin x$, $\cos(2\pi - x) = \cos x$, so $g(2\pi - x) = g(x)$

Queen's rule ($x \to 2\pi - x$ on $[\pi/2, 3\pi/2]$): $I = \int (2\pi - x - \sin x)g\,dx$; adding, $2I = 2\pi \int_{\pi/2}^{3\pi/2} g\,dx$

(Equivalently, with $t = x - \pi$, $(t - \sin t)g$ is odd on $[-\pi/2, \pi/2]$ and integrates to 0)

$\int_{\pi/2}^{3\pi/2} g = \int_{-\pi/2}^{\pi/2} \frac{|\cos t|}{|\sin t|+|\cos t|}dt = 2\int_0^{\pi/2} \frac{\cos t}{\sin t+\cos t}dt$ (even integrand)

King's rule: $\int_0^{\pi/2} \frac{\cos t}{\sin t+\cos t}dt = \int_0^{\pi/2} \frac{\sin t}{\sin t+\cos t}dt = \frac{\pi}{4}$

So $\int g = \frac{\pi}{2}$ and $I = \pi \cdot \frac{\pi}{2} = \frac{\pi^2}{2}$

Correct Option: (A)

Question 03

English

Let $f : [0, 3] \to \mathbb{R}$ be defined by $f(x) = x^{[x]}$ for $x \neq 0$ and $f(0) = 1$, where $[t]$ denotes the greatest integer less than or equal to t. Then $\int_0^3 f(x)\,dx$ is equal to

(A) $\frac{47}{6}$

(B) $\frac{53}{6}$

(C) $\frac{26}{3}$

(D) 9

Only one option is correct.

हिन्दी

माना कि $f : [0, 3] \to \mathbb{R}$, $x \neq 0$ के लिए $f(x) = x^{[x]}$ तथा $f(0) = 1$ द्वारा परिभाषित है, जहाँ $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब $\displaystyle\int_0^3 f(x)\,dx$ बराबर है

(A) $\dfrac{47}{6}$

(B) $\dfrac{53}{6}$

(C) $\dfrac{26}{3}$

(D) 9

केवल एक विकल्प सही है।

Solution

$[x]$ is constant on $[0, 1)$, $[1, 2)$, $[2, 3)$, so split the integral there (the jumps at $1, 2$ do not affect the integral)

$[0, 1)$: $f = x^0 = 1$, integral 1

$[1, 2)$: $f = x$, integral $\frac{4-1}{2} = \frac{3}{2}$

$[2, 3)$: $f = x^2$, integral $\frac{27-8}{3} = \frac{19}{3}$

Total $= 1 + \frac{3}{2} + \frac{19}{3} = \frac{53}{6}$

Correct Option: (B)

Question 04

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x - \dfrac{\sin(\pi x)}{2\pi}$. Then f is a bijection; let f^{-1} denote its inverse. Then the area of the region bounded by the curves $y = f(x)$ and $y = f^{-1}(x)$ and lying in the strip $0 \leq x \leq 3$ is

(A) $\dfrac{3}{\pi^2}$

(B) $\dfrac{3}{\pi}$

(C) $\dfrac{6}{\pi}$

(D) $\dfrac{6}{\pi^2}$

Only one option is correct.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x - \dfrac{\sin(\pi x)}{2\pi}$ द्वारा परिभाषित है। तब f एकैकी आच्छादक है; माना f^{-1} इसका प्रतिलोम फलन है। तब वक्रों $y = f(x)$ तथा $y = f^{-1}(x)$ से परिबद्ध तथा पट्टी $0 \leq x \leq 3$ में स्थित क्षेत्र का क्षेत्रफल है

(A) $\dfrac{3}{\pi^2}$

(B) $\dfrac{3}{\pi}$

(C) $\dfrac{6}{\pi}$

(D) $\dfrac{6}{\pi^2}$

केवल एक विकल्प सही है।

Solution

$f'(x) = 1 - \frac{1}{2}\cos \pi x \geq \frac{1}{2} > 0$: f increasing, so f and f^{-1} meet only on $y = x$, where $\sin \pi x = 0$, i.e. at $x = 0, 1, 2, 3$

The curves are mirror images in $y = x$, so the region consists of three loops, each of area $2\int |x - f(x)|dx$ over a unit interval

Check with the inverse-pair formula on $[0, 1]$: $\int_0^1 f^{-1}(y)dy = 1 \cdot f(1) - 0 - \int_0^1 f = 1 - \left(\frac{1}{2} - \frac{1}{\pi^2}\right)$, so $\int_0^1 (f^{-1} - f) = \frac{2}{\pi^2}$

Each loop: $2\int_0^1 \frac{\sin \pi x}{2\pi} dx = \frac{2}{\pi^2}$ (same for $[1, 2]$, $[2, 3]$ with $|\sin \pi x|$)

Area $= 3 \cdot \frac{2}{\pi^2} = \frac{6}{\pi^2}$

Correct Option: (D)

Question 05

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that

$$f(x) = x^2 + \int_0^x e^{x-t} f(t)\, dt \quad \text{for all } x \in \mathbb{R}.$$

Then $f(1)$ is equal to

(A) $\dfrac{e^2 - 1}{4}$

(B) $\dfrac{e^2 + 1}{4}$

(C) $\dfrac{e^2 - 1}{2}$

(D) $\dfrac{e^2 - 3}{4}$

Only one option is correct.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक सतत फलन है जिसके लिए

$$f(x) = x^2 + \int_0^x e^{x-t} f(t)\, dt, \quad x \in \mathbb{R}.$$

तब $f(1)$ बराबर है

(A) $\dfrac{e^2 - 1}{4}$

(B) $\dfrac{e^2 + 1}{4}$

(C) $\dfrac{e^2 - 1}{2}$

(D) $\dfrac{e^2 - 3}{4}$

केवल एक विकल्प सही है।

Solution

$f(x) = x^2 + e^x \int_0^x e^{-t} f(t)dt$, so $f(0) = 0$; f is differentiable (right side is)
Newton-Leibniz: $f' = 2x + e^x \int_0^x e^{-t} f\, dt + e^x e^{-x} f = 2x + (f - x^2) + f$
$f' - 2f = 2x - x^2$ (linear DE). Particular solution $\frac{x^2}{2} - \frac{x}{2} - \frac{1}{4}$
$f = \frac{x^2}{2} - \frac{x}{2} - \frac{1}{4} + Ce^{2x}$, $f(0) = 0 \Rightarrow C = \frac{1}{4}$
$f(1) = -\frac{1}{4} + \frac{e^2}{4} = \frac{e^2-1}{4}$

Correct Option: (A)

Question 06

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $f(x + y) = f(x) + f(y) + xy(x + y)$ for all $x, y \in \mathbb{R}$ and $\lim\limits_{x\to 0} \dfrac{f(x)}{\sin x} = -3$. Let $G : \mathbb{R} \to \mathbb{R}$ be defined by

$$G(x) = \int_x^{2x} f(t)\, dt.$$

Then the minimum value of G is

(A) $-\dfrac{81}{40}$
(B) $-\dfrac{81}{20}$
(C) $-\dfrac{27}{4}$
(D) $-\dfrac{9}{5}$

Only one option is correct.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए सभी $x, y \in \mathbb{R}$ के लिए $f(x+y) = f(x)+f(y)+xy(x+y)$ तथा $\lim\limits_{x\to 0} \dfrac{f(x)}{\sin x} = -3$ है। माना कि $G : \mathbb{R} \to \mathbb{R}$

$$G(x) = \int_x^{2x} f(t)\, dt$$

द्वारा परिभाषित है। तब G का न्यूनतम मान है

(A) $-\dfrac{81}{40}$
(B) $-\dfrac{81}{20}$
(C) $-\dfrac{27}{4}$
(D) $-\dfrac{9}{5}$

केवल एक विकल्प सही है।

Solution

$x = y = 0$: $f(0) = 0$. Then $\lim \frac{f(x)}{\sin x} = \lim \frac{f(x)-f(0)}{x} = f'(0) = -3$
$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim \left(\frac{f(h)}{h} + x^2 + xh \right) = -3 + x^2$, so $f(x) = \frac{x^3}{3} - 3x$
Newton-Leibniz: $G'(x) = 2f(2x) - f(x) = \frac{16x^3}{3} - 12x - \frac{x^3}{3} + 3x = 5x^3 - 9x$

$G' = 0$ at $x = 0, \pm\frac{3}{\sqrt{5}}$; G' changes $- \to +$ at $\pm\frac{3}{\sqrt{5}}$: minima

$G(x) = \left[\frac{t^4}{12} - \frac{3t^2}{2}\right]_x^{2x} = \frac{5}{4}x^4 - \frac{9}{2}x^2$; at $x^2 = \frac{9}{5}$: $\frac{81}{20} - \frac{81}{10} = -\frac{81}{20}$

Correct Option: (B)

Question 07

English

For each non-negative integer n, let $I_n = \displaystyle\int_0^\infty x^n e^{-4x}\,dx$. Then the value of $\displaystyle\sum_{n=0}^{\infty} \frac{2^n I_n}{n!}$ is

(A) $\frac{1}{4}$
(B) 1
(C) 2
(D) $\frac{1}{2}$

Only one option is correct.

हिन्दी

प्रत्येक अऋणात्मक पूर्णांक n के लिए माना $I_n = \displaystyle\int_0^\infty x^n e^{-4x}\,dx$ है। तब $\displaystyle\sum_{n=0}^{\infty} \frac{2^n I_n}{n!}$ का मान है

(A) $\frac{1}{4}$
(B) 1
(C) 2
(D) $\frac{1}{2}$

केवल एक विकल्प सही है।

Solution

Gamma shortcut: $\int_0^\infty x^n e^{-ax}dx = \frac{\Gamma(n+1)}{a^{n+1}} = \frac{n!}{a^{n+1}}$, so $I_n = \frac{n!}{4^{n+1}}$

$\frac{2^n I_n}{n!} = \frac{2^n}{4^{n+1}} = \frac{1}{4}\left(\frac{1}{2}\right)^n$

Sum $= \frac{1}{4} \cdot \frac{1}{1-\frac{1}{2}} = \frac{1}{2}$

Correct Option: (D)

Question 08

English

Let $f : (0, \infty) \to \mathbb{R}$ be a function, differentiable at $x = 1$, such that $f(xy) = f(x) + f(y)$ for all $x, y \in (0, \infty)$ and $\lim\limits_{h \to 0} \dfrac{f(1+h)}{h} = 2$. Then

$$\lim_{x \to 1} \frac{1}{(x-1)^2} \int_x^{x^2} f(t)\,dt$$

is equal to

(A) 6

(B) 2
(C) 3
(D) $\frac{3}{2}$

Only one option is correct.

हिन्दी

माना कि $f : (0, \infty) \to \mathbb{R}$ एक फलन है जो $x = 1$ पर अवकलनीय है, जिसके लिए सभी $x, y \in (0, \infty)$ के लिए $f(xy) = f(x) + f(y)$ तथा $\lim\limits_{h \to 0} \frac{f(1+h)}{h} = 2$ है। तब

$$\lim_{x \to 1} \frac{1}{(x-1)^2} \int_x^{x^2} f(t)\, dt$$

बराबर है

(A) 6
(B) 2
(C) 3
(D) $\frac{3}{2}$

केवल एक विकल्प सही है।

Solution

$x = y = 1$: $f(1) = 0$, so $f'(1) = \lim \frac{f(1+h)}{h} = 2$

$f'(x) = \lim \frac{f(x(1+h/x)) - f(x)}{h} = \lim \frac{f(1+h/x)}{h} = \frac{f'(1)}{x} = \frac{2}{x}$, so $f(x) = 2 \ln x$

Limit is $\frac{0}{0}$; L'Hospital with Newton-Leibniz: $\frac{2xf(x^2) - f(x)}{2(x-1)} = \frac{(8x-2)\ln x}{2(x-1)}$

$\frac{\ln x}{x-1} \to 1$, so the limit is $\frac{8-2}{2} = 3$

Correct Option: (C)

Question 09

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a polynomial function such that

$$\int_{x-1}^{x+1} f(t)\, dt = x^3 \quad \text{for all } x \in \mathbb{R}.$$

Then the value of $f(-4)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक बहुपद फलन है जिसके लिए

$$\int_{x-1}^{x+1} f(t)\, dt = x^3, \quad x \in \mathbb{R}.$$

तब $f(-4)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Newton-Leibniz: $f(x+1) - f(x-1) = 3x^2$ for all x

$f(x+1) - f(x-1)$ lowers the degree by one, so f is a cubic $ax^3 + bx^2 + cx + d$:

$a(6x^2 + 2) + 4bx + 2c = 3x^2 \Rightarrow a = \frac{1}{2},\ b = 0,\ c = -\frac{1}{2}$

d is not fixed by the derivative; FTC part 2 at $x = 0$: $\int_{-1}^{1} f = 2d = 0$ (odd terms vanish) $\Rightarrow d = 0$

$f(x) = \frac{x^3 - x}{2}$, $f(-4) = \frac{-64+4}{2} = -30$

Final Answer: -30

Question 10

English

Let k be the positive integer for which

$$\int_0^{k\pi/2} \frac{dx}{\sin^2 x + 9\cos^2 x} = \frac{7\pi}{2}.$$

Then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि k वह धनात्मक पूर्णांक है जिसके लिए

$$\int_0^{k\pi/2} \frac{dx}{\sin^2 x + 9\cos^2 x} = \frac{7\pi}{2}.$$

तब k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

With $t = \tan x$: an antiderivative is $F(x) = \frac{1}{3}\tan^{-1}\left(\frac{\tan x}{3}\right)$, but F jumps at $x = \frac{\pi}{2} + m\pi$, so it may be used only on intervals free of these points

On $[0, \frac{\pi}{2})$: $\int_0^{\pi/2} = \lim_{x\to\pi/2^-} F(x) - F(0) = \frac{1}{3} \cdot \frac{\pi}{2} = \frac{\pi}{6}$

The integrand g has period π and $g(\pi - x) = g(x)$, so every interval $[\frac{m\pi}{2}, \frac{(m+1)\pi}{2}]$ contributes $\frac{\pi}{6}$

$k \cdot \frac{\pi}{6} = \frac{7\pi}{2} \Rightarrow k = 21$ (the naive $F(\frac{k\pi}{2}) - F(0)$ gives 0 or $\frac{\pi}{6}$)

Final Answer: 21

Question 11

English

Let $F : \mathbb{R} \to \mathbb{R}$ be defined by

$$F(x) = \int_0^x \frac{1 + t^2}{1 - t^2 + t^4}\, dt.$$

Then the value of $F\left(-\frac{1+\sqrt{5}}{2}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : \mathbb{R} \to \mathbb{R}$

$$F(x) = \int_0^x \frac{1+t^2}{1-t^2+t^4}\,dt$$

द्वारा परिभाषित है। तब $F\left(-\frac{1+\sqrt{5}}{2}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$1-t^2+t^4 > 0$, so the integrand is continuous and F is continuous and odd with $F' = \frac{1+t^2}{1-t^2+t^4}$ (FTC)

$\frac{d}{dx}\tan^{-1}\frac{x}{1-x^2} = \frac{1+x^2}{1-x^2+x^4}$ for $x \neq \pm 1$, but $\tan^{-1}\frac{x}{1-x^2}$ jumps by π at $x = \pm 1$

So $F(x) = \tan^{-1}\frac{x}{1-x^2}$ on $(-1,1)$ and $F(x) = \tan^{-1}\frac{x}{1-x^2} - \pi$ for $x < -1$ (continuity at -1: left limit of $\tan^{-1}$ is $\frac{\pi}{2}$, $F(-1) = -\frac{\pi}{2}$)

$x_0 = -\frac{1+\sqrt{5}}{2}$ satisfies $x_0^2 + x_0 - 1 = 0$, so $\frac{x_0}{1-x_0^2} = 1$

$F(x_0) = \frac{\pi}{4} - \pi = -\frac{3\pi}{4} \approx -2.36$

Final Answer: -2.36

Question 12

English

The value of $\displaystyle\int_{-1}^{3} \frac{|x|}{(1+x^2)^2}\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\displaystyle\int_{-1}^{3} \frac{|x|}{(1+x^2)^2}\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The substitution $u = x^2$ is not one-one on $[-1,3]$; it is monotonic on $[-1,0]$ and on $[0,3]$ separately

$[-1,0]$: $|x| = -x$, $u = x^2$ decreasing from 1 to 0: $\int_{-1}^{0} \frac{-x\,dx}{(1+x^2)^2} = \frac{1}{2}\int_0^1 \frac{du}{(1+u)^2} = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$

$[0,3]$: $\int_0^3 \frac{x\,dx}{(1+x^2)^2} = \frac{1}{2}\int_0^9 \frac{du}{(1+u)^2} = \frac{1}{2}\cdot\frac{9}{10} = \frac{9}{20}$

Total $= \frac{1}{4} + \frac{9}{20} = \frac{7}{10}$ (one substitution over the whole interval would give the wrong value 0.20)

Final Answer: 0.70

Question 13

English

The value of $\displaystyle\int_{1}^{e^{12}} \frac{x\,\log_e x\,(12-\log_e x)}{x^2+e^{12}}\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\int_1^{e^{12}} \frac{x \log_e x \ (12 - \log_e x)}{x^2 + e^{12}}\, dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Substitute $x = e^t$ (increasing, so the limits become $t = 0$ to $t = 12$): $dx = e^t dt$, $\frac{x\,dx}{x^2+e^{12}} = \frac{e^{2t}dt}{e^{2t}+e^{12}}$

$I = \int_0^{12} t(12-t)\,h(t)\,dt$ with $h(t) = \frac{1}{1+e^{12-2t}}$

King's rule $t \to 12 - t$: $t(12-t)$ unchanged, $h(12-t) = \frac{1}{1+e^{2t-12}} = 1 - h(t)$

$2I = \int_0^{12} t(12-t)\,dt = 864 - 576 = 288 \Rightarrow I = 144$

Final Answer: 144

Question 14

English

Let $f : [0, 4] \to \mathbb{R}$ be defined by

$$f(x) = \frac{(x^2 - 4x + 5)\sqrt{x}}{\sqrt{x} + \sqrt{4 - x}}.$$

Let $I_1 = \int_0^4 f(x)\,dx$ and $I_2 = \int_1^3 f(x)\,dx$. Then the value of $\frac{I_1}{I_2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : [0, 4] \to \mathbb{R}$

$$f(x) = \frac{(x^2 - 4x + 5)\sqrt{x}}{\sqrt{x} + \sqrt{4 - x}}$$

द्वारा परिभाषित है। माना $I_1 = \int_0^4 f(x)\,dx$ तथा $I_2 = \int_1^3 f(x)\,dx$ है। तब $\frac{I_1}{I_2}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$w(x) = x^2 - 4x + 5 = (x-2)^2 + 1$ satisfies $w(4-x) = w(x)$; $h(x) = \frac{\sqrt{x}}{\sqrt{x}+\sqrt{4-x}}$ satisfies $h(4-x) = 1 - h(x)$

King's rule on $[0, 4]$: $I_1 = \int_0^4 w(1-h)$, so $2I_1 = \int_0^4 w = \frac{16}{3} + 4 = \frac{28}{3}$, $I_1 = \frac{14}{3}$

Queen's rule on $[1, 3]$ $(a + b = 4)$: $2I_2 = \int_1^3 w = \frac{2}{3} + 2 = \frac{8}{3}$, $I_2 = \frac{4}{3}$

$\frac{I_1}{I_2} = \frac{14}{4} = 3.5$

Final Answer: 3.50

Question 15

English

The value of

$$\int_{-\pi/2}^{3\pi/2} \frac{\sin^4 x}{\left(\sin^4 x + \cos^4 x\right)\left(1 + e^{\cos x}\right)}\, dx$$

is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$$\int_{-\pi/2}^{3\pi/2} \frac{\sin^4 x}{\left(\sin^4 x + \cos^4 x\right)\left(1 + e^{\cos x}\right)}\, dx$$

का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $g(x) = \frac{\sin^4 x}{\sin^4 x + \cos^4 x}$. Queen's rule $x \to \pi - x$ on $[-\frac{\pi}{2}, \frac{3\pi}{2}]$: g unchanged, $\cos x \to -\cos x$

$\frac{1}{1+e^{-\cos x}} = 1 - \frac{1}{1+e^{\cos x}}$, so $2I = \int_{-\pi/2}^{3\pi/2} g\, dx$

Split: $\int_{-\pi/2}^{0} g = \int_{0}^{\pi/2} g$ (g even); $\int_{0}^{\pi} g = 2\int_{0}^{\pi/2} g$ ($g(\pi - x) = g(x)$); $\int_{\pi}^{3\pi/2} g = \int_{-\pi/2}^{0} g$ ($x \to \pi - x$)

So $\int_{-\pi/2}^{3\pi/2} g = 4\int_{0}^{\pi/2} g$, and by King's rule $\int_{0}^{\pi/2} g = \frac{\pi}{4}$ ($g(\frac{\pi}{2} - x) = 1 - g(x)$)

$2I = \pi \Rightarrow I = \frac{\pi}{2} \approx 1.57$

Final Answer: 1.57

Question 16

English

Considering the principal value of $\cot^{-1}$, which takes values in $(0, \pi)$, let

$$L = \lim_{n\to\infty} \sum_{k=1}^{2n} \frac{k(k-2n)}{n^3} \cot^{-1}\left(\frac{(k-n)^3}{n^3}\right).$$

Then the value of L is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\cot^{-1}$ के मुख्य मान (जो $(0, \pi)$ में है) को लेते हुए माना

$$L = \lim_{n\to\infty} \sum_{k=1}^{2n} \frac{k(k-2n)}{n^3} \cot^{-1}\left(\frac{(k-n)^3}{n^3}\right).$$

तब L का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

With $x=\frac{k}{n}$, $\Delta x=\frac{1}{n}$: $\frac{k(k-2n)}{n^3}=\frac{1}{n}\cdot x(x-2)$, $k=1..2n$ covers $(0,2]$: $L=\int_0^2 x(x-2)\cot^{-1}\left((x-1)^3\right)dx$
King's rule $x\to 2-x$: $x(x-2)$ unchanged, $(x-1)^3\to -(x-1)^3$ (odd), $\cot^{-1}(-y)=\pi-\cot^{-1}y$
$2L=\pi\int_0^2 x(x-2)\,dx=\pi\left(\frac{8}{3}-4\right)=-\frac{4\pi}{3}$
$L=-\frac{2\pi}{3}\approx -2.09$

Final Answer: -2.09

Question 17

English

Let

$$\int_{\pi/8}^{3\pi/8}\frac{dx}{\sin^4 x\,\cos^4 x\,(1+\tan^5 x)}=\frac{p}{q},$$

where p and q are positive integers with $\gcd(p,q)=1$. Then the value of $p+q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना

$$\int_{\pi/8}^{3\pi/8}\frac{dx}{\sin^4 x\,\cos^4 x\,(1+\tan^5 x)}=\frac{p}{q}$$

है, जहाँ p तथा q धनात्मक पूर्णांक हैं तथा $\gcd(p,q)=1$ है। तब $p+q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Queen's rule $x\to\frac{\pi}{2}-x$ (here $a+b=\frac{\pi}{2}$): $\sin^4 x\cos^4 x$ unchanged, $\frac{1}{1+\tan^5 x}\to\frac{1}{1+\cot^5 x}=\frac{\tan^5 x}{1+\tan^5 x}$
Adding: $2I=\int_{\pi/8}^{3\pi/8}\frac{dx}{\sin^4 x\cos^4 x}=16\int_{\pi/8}^{3\pi/8}\csc^4 2x\,dx$
$u=2x$: $I=4\int_{\pi/4}^{3\pi/4}\csc^4 u\,du$; $\csc^4 u=(1+\cot^2 u)\csc^2 u$, antiderivative $-\cot u-\frac{\cot^3 u}{3}$
$I=4\left[\left(1+\frac{1}{3}\right)-\left(-1-\frac{1}{3}\right)\right]=4\cdot\frac{8}{3}=\frac{32}{3}$, so $p+q=35$

Final Answer: 35

Question 18

English

Let $f:[-8,8]\to\mathbb{R}$ be defined by $f(x)=x^5\cos x+|x|\,(8-|x|)\,\big|4-|x|\big|$. Then the value of $\int_{-8}^{8}f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f:[-8,8]\to\mathbb{R}$, $f(x)=x^5\cos x+|x|\,(8-|x|)\,\big|4-|x|\big|$ द्वारा परिभाषित है। तब $\int_{-8}^{8}f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$x^5 \cos x$ is odd: its integral over $[-8, 8]$ is 0. The other term $E(x)$ is even: $\int_{-8}^{8} E = 2\int_0^8 x(8-x)|4-x|\,dx$

On $[0, 8]$, $g(x) = x(8-x)|4-x|$ satisfies $g(8-x) = g(x)$, so $\int_0^8 g = 2\int_0^4 x(8-x)(4-x)dx$

$\int_0^4 (32x - 12x^2 + x^3)dx = 256 - 256 + 64 = 64$

Integral $= 2 \cdot 2 \cdot 64 = 256$

Final Answer: 256

Question 19

English

The value of $\displaystyle\int_0^4 \left(x^2 - 4x\right)\left|x^2 - 4x + 3\right| \, dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\displaystyle\int_0^4 \left(x^2 - 4x\right)\left|x^2 - 4x + 3\right| \, dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Both factors depend on $x^2 - 4x = (x-2)^2 - 4$, so $f(4-x) = f(x)$: $\int_0^4 f = 2\int_0^2 f$

Let $q = x^2 - 4x + 3 = (x-1)(x-3)$: $q \geq 0$ on $[0, 1]$, $q \leq 0$ on $[1, 2]$; $f = (q-3)|q|$

$\int_0^1 q = \frac{4}{3}$, $\int_1^2 q = -\frac{2}{3}$, $\int_0^1 q^2 = \frac{38}{15}$, $\int_1^2 q^2 = \frac{8}{15}$

$\int_0^2 f = \left(\frac{38}{15} - 4\right) - \left(\frac{8}{15} + 2\right) = 2 - 6 = -4$

Integral $= 2(-4) = -8$

Final Answer: -8

Question 20

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \dfrac{\sin^2 x}{1 + \cos^2 x}$. Then the value of $\lim\limits_{T\to\infty} \dfrac{1}{T} \displaystyle\int_0^T f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{\sin^2 x}{1 + \cos^2 x}$ द्वारा परिभाषित है। तब $\lim\limits_{T\to\infty} \dfrac{1}{T} \displaystyle\int_0^T f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

f has period π; for $n\pi \le T < (n+1)\pi$: $nA \le \int_0^T f \le (n+1)A$ with $A = \int_0^\pi f$, so the limit is $\frac{A}{\pi}$

$f(\pi - x) = f(x)$, so $A = 2\int_0^{\pi/2} f$ (this also keeps $\tan x$ continuous on $[0, \frac{\pi}{2})$)

$f = \frac{2}{1+\cos^2 x} - 1$; with $t = \tan x$: $\int_0^{\pi/2} \frac{dx}{1+\cos^2 x} = \int_0^\infty \frac{dt}{t^2+2} = \frac{\pi}{2\sqrt{2}}$

$\int_0^{\pi/2} f = \frac{\pi}{\sqrt{2}} - \frac{\pi}{2}$, $A = \pi(\sqrt{2} - 1)$, limit $= \sqrt{2} - 1 \approx 0.41$

Final Answer: 0.41

Question 21

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that $f(x+1) = -f(x)$ for all $x \in \mathbb{R}$ and $\displaystyle\int_0^1 f(x)\,dx = 2$. Then the value of $\displaystyle\int_0^{20} x\,f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक सतत फलन है जिसके लिए सभी $x \in \mathbb{R}$ के लिए $f(x+1) = -f(x)$ तथा $\displaystyle\int_0^1 f(x)\,dx = 2$ है। तब $\displaystyle\int_0^{20} x\,f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$f(x+2) = -f(x+1) = f(x)$: f has period 2; let $B = \int_0^1 tf(t)\,dt$

$\int_0^2 f = \int_0^1 f + \int_0^1 f(t+1)dt = 2 - 2 = 0$

On $[2m, 2m+2]$, $x = t + 2m$: $\int_{2m}^{2m+2} xf(x)dx = \int_0^2 (t+2m)f(t)dt = \int_0^2 tf(t)dt + 2m \cdot 0$

$\int_0^2 tf = B + \int_0^1 (t+1)(-f(t))dt = B - B - 2 = -2$

Ten periods in $[0, 20]$: integral $= 10(-2) = -20$

Final Answer: -20

Question 22

English

Let $f : \mathbb{R} \setminus \{1, 2\} \to \mathbb{R}$ be defined by $f(x) = \dfrac{x^3 - 3x^2 + 2x}{|x^2 - 3x + 2|}$. Then the value of $\displaystyle\int_{-3}^{2} f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \setminus \{1, 2\} \to \mathbb{R}$, $f(x) = \dfrac{x^3 - 3x^2 + 2x}{|x^2 - 3x + 2|}$ द्वारा परिभाषित है। तब $\displaystyle\int_{-3}^{2} f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$x^3 - 3x^2 + 2x = x(x-1)(x-2)$, so $f(x) = x \cdot \operatorname{sgn}\big((x-1)(x-2)\big)$: $f = x$ for $x < 1$, $f = -x$ for $1 < x < 2$

f is bounded with a jump at $x = 1$, so split there: $\int_{-3}^{1} x\,dx = \frac{1-9}{2} = -4$, $\int_{1}^{2}(-x)dx = -\frac{3}{2}$

Total $= -4 - 1.5 = -5.5$

Final Answer: -5.50

Question 23

English

Let $f : [0, 14] \to \mathbb{R}$ be defined by $f(x) = \max\{[x], |x - 7|\}$, where $[t]$ denotes the greatest integer less than or equal to t. Then the value of $\displaystyle\int_{0}^{14} f(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : [0, 14] \to \mathbb{R}$, $f(x) = \max\{[x], |x - 7|\}$ द्वारा परिभाषित है, जहाँ $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब $\displaystyle\int_{0}^{14} f(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

On $[n, n+1)$, $[x] = n$; compare n with $|x - 7|$

$n = 0, 1, 2, 3$ $(x < 4)$: $|x - 7| = 7 - x > 3 \geq n$, so $f = 7 - x$: $\int_0^4 (7 - x)dx = 28 - 8 = 20$

$n = 4, 5, 6$ $(4 \leq x < 7)$: $7 - x \leq 3 < n$, so $f = n$: $4 + 5 + 6 = 15$

$n = 7, \ldots, 13$: $|x - 7| = x - 7 < n - 6 < n$, so $f = n$: $7 + 8 + \cdots + 13 = 70$

Total $= 20 + 15 + 70 = 105$

Final Answer: 105

Question 24

English

For a real number t, let $\{t\} = t - [t]$, where $[t]$ denotes the greatest integer less than or equal to t. Then the value of $\displaystyle\int_{1/4}^{1} \left\{\frac{1}{x}\right\} dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्या t के लिए माना $\{t\} = t - [t]$, जहाँ $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब $\displaystyle\int_{1/4}^{1} \left\{\frac{1}{x}\right\} dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\frac{1}{x} \in (n, n+1]$ for $x \in \left[\frac{1}{n+1}, \frac{1}{n}\right)$; on such an interval $\left\{\frac{1}{x}\right\} = \frac{1}{x} - n$ (and $= 0$ at $x = \frac{1}{n}$)

$\int_{1/(n+1)}^{1/n} \left(\frac{1}{x} - n\right) dx = \ln\frac{n+1}{n} - \frac{1}{n+1}$

Sum for $n = 1, 2, 3$: $\ln 4 - \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) = 2\ln 2 - \frac{13}{12} \approx 1.3863 - 1.0833 = 0.30$

Final Answer: 0.30

Question 25

English

Let $f : [0, 5] \to \mathbb{R}$ be defined by $f(x) = \min\left\{\{x\}, \dfrac{|x-3|}{3}\right\}$, where $\{t\} = t - [t]$ and $[t]$ denotes the greatest integer less than or equal to t. Then the value of $\displaystyle\int_0^5 f(x)\, dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : [0, 5] \to \mathbb{R}$, $f(x) = \min\left\{\{x\}, \dfrac{|x-3|}{3}\right\}$ द्वारा परिभाषित है, जहाँ $\{t\} = t - [t]$ तथा $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब $\displaystyle\int_0^5 f(x)\, dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

On $[n, n+1)$, $\{x\} = x - n$; the line $\frac{|x-3|}{3}$ meets it at $x = \frac{3}{4}, \frac{3}{2}, \frac{9}{4}$ (left of 3) and $\frac{9}{2}$ (right of 3)

$[0, 1]$: $\int_0^{3/4} x\, dx + \int_{3/4}^1 \left(1 - \frac{x}{3}\right) dx = \frac{9}{32} + \frac{17}{96} = \frac{11}{24}$

$[1, 2]$: $\int_1^{3/2} (x-1) dx + \int_{3/2}^2 \left(1 - \frac{x}{3}\right) dx = \frac{1}{8} + \frac{5}{24} = \frac{1}{3}$

$[2, 3]$: $\int_2^{9/4} (x-2) dx + \int_{9/4}^3 \left(1 - \frac{x}{3}\right) dx = \frac{1}{32} + \frac{3}{32} = \frac{1}{8}$

$[3, 4]$: $\frac{x-3}{3} \le x - 3$, so $\int_3^4 \frac{x-3}{3} dx = \frac{1}{6}$; $[4, 5]$: $\int_4^{9/2} (x-4) dx + \int_{9/2}^5 \frac{x-3}{3} dx = \frac{1}{8} + \frac{7}{24} = \frac{5}{12}$

Total $= \frac{11+8+3+4+10}{24} = \frac{36}{24} = 1.5$

Final Answer: 1.50

Question 26

English

Let $g : \mathbb{R} \to \mathbb{R}$ be the function such that $(g(x))^3 + g(x) = x$ for all $x \in \mathbb{R}$. Then the value of $\int_2^{30} g(x)\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $g : \mathbb{R} \to \mathbb{R}$ वह फलन है जिसके लिए सभी $x \in \mathbb{R}$ के लिए $(g(x))^3 + g(x) = x$ है। तब $\int_2^{30} g(x)\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$h(t) = t^3 + t$ is increasing and onto $\mathbb{R}$; $h(g(x)) = x$ means $g = h^{-1}$

$g(2) = 1$, $g(30) = 3$ (since $h(1) = 2$, $h(3) = 30$)

$\int_2^{30} h^{-1}(y)dy + \int_1^3 h(t)dt = 3 \cdot 30 - 1 \cdot 2 = 88$

$\int_1^3 (t^3 + t)dt = \frac{81-1}{4} + \frac{9-1}{2} = 24$, so the integral $= 88 - 24 = 64$

Final Answer: 64

Question 27

English

Let $F : \mathbb{R} \to \mathbb{R}$ be defined by $F(x) = \int_x^{x+1} t^2\, 2^{-t}\,dt$. If F has a local maximum at $x = \alpha$, then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : \mathbb{R} \to \mathbb{R}$, $F(x) = \int_x^{x+1} t^2\, 2^{-t}\,dt$ द्वारा परिभाषित है। यदि $x = \alpha$ पर F का स्थानीय अधिकतम है, तो α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Newton-Leibniz: $F'(x) = (x+1)^2 2^{-x-1} - x^2 2^{-x} = 2^{-x-1}\left[(x+1)^2 - 2x^2\right]$

$(x+1)^2 - 2x^2 = -(x^2 - 2x - 1)$, zero at $x = 1 \pm \sqrt{2}$

$F' < 0$ for $x < 1 - \sqrt{2}$, > 0 between the roots, < 0 for $x > 1 + \sqrt{2}$: local min at $1 - \sqrt{2}$, local max at $1 + \sqrt{2}$

$\alpha = 1 + \sqrt{2} \approx 2.41$

Final Answer: 2.41

Question 28

English

Let $F : \mathbb{R} \to \mathbb{R}$ be defined by $F(x) = \int_{-2}^{x} (t-3)\,|t-2|\,dt$. Then the minimum value of F is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : \mathbb{R} \to \mathbb{R}$, $F(x) = \int_{-2}^{x} (t-3)\,|t-2|\,dt$ द्वारा परिभाषित है। तब F का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$F'(x) = (x-3)|x-2|$: zero at $x = 2$ and $x = 3$

$F' < 0$ on $(-\infty, 2)$ and $(2, 3)$ (no sign change at 2), $F' > 0$ on $(3, \infty)$: F decreases up to 3, then increases, so the minimum is $F(3)$

$F(3) = \int_{-2}^{2}(t-3)(2-t)dt + \int_{2}^{3}(t-3)(t-2)dt = -\frac{88}{3} - \frac{1}{6} = -\frac{59}{2}$

Final Answer: -29.50

Question 29

English

Let $G : \mathbb{R} \to \mathbb{R}$ be defined by

$$G(x) = \int_{0}^{x^2+2x} (t+1)^2(t-3)\,e^{t^2}\,dt.$$

Then the sum of all the points of local extremum of G is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $G : \mathbb{R} \to \mathbb{R}$

$$G(x) = \int_{0}^{x^2+2x} (t+1)^2(t-3)\,e^{t^2}\,dt$$

द्वारा परिभाषित है। तब G के सभी स्थानीय चरम बिन्दुओं का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Newton-Leibniz with $u = x^2 + 2x$: $G'(x) = (2x+2)(u+1)^2(u-3)e^{u^2}$

$u + 1 = (x+1)^2$, $u - 3 = (x+3)(x-1)$: $G'(x) = 2(x+1)^5(x+3)(x-1)e^{u^2}$

Critical points $-3, -1, 1$. Signs: $-$ on $(-\infty, -3)$, $+$ on $(-3, -1)$, $-$ on $(-1, 1)$, $+$ on $(1, \infty)$

Minima at $-3, 1$; maximum at -1 (odd power 5: sign changes, though $G''(-1) = 0$, so the second derivative test fails there)

Sum $= -3 - 1 + 1 = -3$

Final Answer: -3

Question 30

English

The value of $\lim\limits_{n\to\infty} \dfrac{1}{n^3} \sum\limits_{k=1}^{6n} \left|k^2 - 9n^2\right|$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\lim\limits_{n\to\infty} \dfrac{1}{n^3} \sum\limits_{k=1}^{6n} \left|k^2 - 9n^2\right|$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\frac{1}{n^3}|k^2 - 9n^2| = \frac{1}{n}\left|\left(\frac{k}{n}\right)^2 - 9\right|$; $\frac{k}{n}$ runs over $(0, 6]$ with step $\frac{1}{n}$

Limit $= \int_0^6 |x^2 - 9|\, dx = \int_0^3 (9 - x^2)dx + \int_3^6 (x^2 - 9)dx = 18 + 36 = 54$

Final Answer: 54

Question 31

English

The value of $\lim\limits_{n\to\infty} \left(\prod\limits_{k=1}^{n} \dfrac{2k}{n+k}\right)^{1/n}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\lim\limits_{n\to\infty} \left(\prod\limits_{k=1}^{n} \dfrac{2k}{n+k}\right)^{1/n}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Take $\log_e$: $\log_e L = \lim \frac{1}{n} \sum_{k=1}^{n} \log_e \frac{2k/n}{1+k/n} = \int_0^1 \log_e \frac{2x}{1+x} dx$

$= \log_e 2 + \int_0^1 \log_e x\, dx - \int_0^1 \log_e(1+x)dx = \log_e 2 - 1 - (2\log_e 2 - 1) = -\log_e 2$

$L = e^{-\log_e 2} = \frac{1}{2}$

Final Answer: 0.50

Question 32

English

Let $I = \displaystyle\int_0^{30} \frac{x}{1+e^{-x}}\,dx$. Then the greatest integer less than or equal to I is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $I = \displaystyle\int_0^{30} \frac{x}{1+e^{-x}}\,dx$ है। तब I से कम या उसके बराबर महत्तम पूर्णांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$\frac{x}{1+e^{-x}} = x - \frac{x}{e^x+1}$, so $I = 450 - D$ with $D = \int_0^{30} \frac{x}{e^x+1}dx$

$0 < \frac{x}{e^x+1} < xe^{-x}$ on $(0, 30]$, so $0 < D < \int_0^{30} xe^{-x}dx = 1 - 31e^{-30} < 1$

$449 < I < 450 \Rightarrow [I] = 449$

Final Answer: 449

Question 33

English

Let $I = \displaystyle\int_0^{1} \cos\left(x^2\right)\,dx$. Then the greatest integer less than or equal to $100I$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $I = \displaystyle\int_0^{1} \cos\left(x^2\right)\,dx$ है। तब $100I$ से कम या उसके बराबर महत्तम पूर्णांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $t \geq 0$: $1 - \frac{t^2}{2} \leq \cos t \leq 1 - \frac{t^2}{2} + \frac{t^4}{24}$ (monotonicity, below)

Lower: $\phi(t) = \cos t - 1 + \frac{t^2}{2}$, $\phi' = t - \sin t \geq 0$, $\phi(0) = 0 \Rightarrow \phi \geq 0$

Upper: $\psi(t) = 1 - \frac{t^2}{2} + \frac{t^4}{24} - \cos t$, $\psi' = -t + \frac{t^3}{6} + \sin t \geq 0$ (its derivative $-1 + \frac{t^2}{2} + \cos t = \phi \geq 0$), so $\psi \geq 0$

Put $t = x^2$ and integrate on $[0, 1]$: $1 - \frac{1}{10} \leq I \leq 1 - \frac{1}{10} + \frac{1}{216}$, i.e. $0.9 \leq I \leq 0.9047$

$90 \leq 100I \leq 90.47 \Rightarrow [100I] = 90$

Final Answer: 90

Question 34

English

The value of $\int_0^4 x^2\sqrt{4x-x^2}\,dx$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\int_0^4 x^2\sqrt{4x-x^2}\,dx$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$x = 4\sin^2\theta$, $\theta : 0 \to \frac{\pi}{2}$, $dx = 8\sin\theta\cos\theta\,d\theta$, $\sqrt{4x-x^2} = 4\sin\theta\cos\theta$

Integral $= \int_0^{\pi/2} 16\sin^4\theta \cdot 4\sin\theta\cos\theta \cdot 8\sin\theta\cos\theta\,d\theta = 512\int_0^{\pi/2}\sin^6\theta\cos^2\theta\,d\theta$

Wallis: $\int_0^{\pi/2}\sin^6\cos^2 = \frac{5\cdot3\cdot1\cdot1}{8\cdot6\cdot4\cdot2}\cdot\frac{\pi}{2} = \frac{5\pi}{256}$

Integral $= 512\cdot\frac{5\pi}{256} = 10\pi \approx 31.42$

Final Answer: 31.42

Question 35

English

For a real number $p > -1$, let $W(p) = \int_0^{\pi/2}\sin^p x\,dx$. Then the value of $W\left(\frac{5}{2}\right)W\left(\frac{7}{2}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्या $p > -1$ के लिए माना $W(p) = \int_0^{\pi/2}\sin^p x\,dx$ है। तब $W\left(\frac{5}{2}\right)W\left(\frac{7}{2}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Gamma form: $W(p) = \frac{\sqrt{\pi}}{2}\cdot\frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p}{2}+1\right)}$

$W(p)W(p+1) = \frac{\pi}{4}\cdot\frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p+3}{2}\right)} = \frac{\pi}{4}\cdot\frac{2}{p+1} = \frac{\pi}{2(p+1)}$ (using $\Gamma(s+1) = s\Gamma(s)$)

This is the Wallis relation $(n+1)W_nW_{n+1} = \frac{\pi}{2}$ extended to real p

$p = \frac{5}{2}$: $W\left(\frac{5}{2}\right)W\left(\frac{7}{2}\right) = \frac{\pi}{7} \approx 0.45$

Final Answer: 0.45

Question 36

English

For each non-negative integer n, let $I_n = \int_0^1 (1-x)^n e^x \, dx$. If $I_4 = ae+b$, where a and b are integers, then the value of $a+b$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रत्येक अऋणात्मक पूर्णांक n के लिए माना $I_n = \int_0^1 (1-x)^n e^x \, dx$ है। यदि $I_4 = ae+b$ है, जहाँ a तथा b पूर्णांक हैं, तो $a+b$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By parts ($u = (1-x)^n$, $dv = e^x dx$): $I_n = [(1-x)^n e^x]_0^1 + n\int_0^1 (1-x)^{n-1} e^x dx = -1 + nI_{n-1}$ for $n \geq 1$

$I_0 = e-1$, $I_1 = e-2$, $I_2 = 2e-5$, $I_3 = 6e-16$, $I_4 = 24e-65$

e is irrational, so $a = 24$, $b = -65$ are unique; $a+b = -41$

Final Answer: -41

Question 37

English

Considering the principal value of $\cos^{-1}$, for each non-negative integer n let $I_n = \int_0^1 x^n \cos^{-1} x \, dx$. If $I_6 = \dfrac{p}{q}$, where p and q are positive integers with $\gcd(p,q) = 1$, then the value of $p+q$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\cos^{-1}$ के मुख्य मान को लेते हुए, प्रत्येक अऋणात्मक पूर्णांक n के लिए माना $I_n = \int_0^1 x^n \cos^{-1} x \, dx$ है। यदि $I_6 = \dfrac{p}{q}$ है, जहाँ p तथा q धनात्मक पूर्णांक हैं तथा $\gcd(p,q) = 1$ है, तो $p+q$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By parts: $I_n = \left[\frac{x^{n+1}}{n+1}\cos^{-1} x\right]_0^1 + \frac{1}{n+1}\int_0^1 \frac{x^{n+1}}{\sqrt{1-x^2}}dx = \frac{1}{n+1}J_{n+1}$

$x = \sin\theta$: $J_m = \int_0^{\pi/2} \sin^m\theta \, d\theta$ (Wallis), so $J_m = \frac{m-1}{m}J_{m-2}$ and $I_n = \frac{n(n-1)}{(n+1)^2}I_{n-2}$, $I_0 = 1$

$I_2 = \frac{2}{9}$, $I_4 = \frac{12}{25}\cdot\frac{2}{9} = \frac{8}{75}$, $I_6 = \frac{30}{49}\cdot\frac{8}{75} = \frac{16}{245}$

(Check: $I_6 = \frac{1}{7}J_7 = \frac{1}{7}\cdot\frac{6\cdot4\cdot2}{7\cdot5\cdot3} = \frac{16}{245}$); $p+q = 16+245 = 261$

Final Answer: 261